Maintenance Schedule

Gas engine-generator set
mtu 8V500 GS - B408-ct135
mtu 12V500 GS - B412-ct135
Biogas - interval 2,000 - 64,000 operating hrs
50 Hz / 60 Hz
Application group 3A

MS50303/00E

A Rolls-Royce solution

Master scope of supply name	Sales name	Engine model	kW/cyl.	Application group
GG08VA500A1	mtu 8V500 GS	B408-ct135	45 kW/cyl.	3A, continuous operation, unrestricted
GG08VA500A2	mtu 8V500 GS	B408-ct135	45 kW/cyl.	3A, continuous operation, unrestricted
GG12VA500A1	mtu 12V500 GS	B412-ct135	46 kW/cyl.	3A, continuous operation, unrestricted
GG12VA500A2	mtu 12V500 GS	B412-ct135	46 kW/cyl.	3A, continuous operation, unrestricted

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Additional notes on the maintenance of gas systems

General information

The maintenance schedule matrix normally finishes with extended component maintenance or with engine overhaul. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Important
Repairs tasks on the gas system, for which the gas-conducting area is opened, and leak or stress tests must only be carried out by specialized companies. As a rule these are companies that are approved for the gas industry with qualified personnel who have been trained for technical gas applications.

Important
Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in Fluids and Lubricants Specifications (A001072/..) and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet Rolls-Royce Solutions specifications and be approved by the relevant component manufacturer.

Among other items, the operator/customer must carry out the following additional maintenance work:
- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Gas filter:
  The maintenance interval depends on gaseous fuel quality (purity). Gaseous fuel filters upstream of the engine must be regularly cleaned and renewed if necessary.
- Battery:
  Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

Items listed in this maintenance schedule but which are not included in the scope of delivery may be disregarded.

Generator circuit breaker maintenance

Please refer to the Operating Instructions provided by the manufacturer for the intervals for these maintenance tasks.

Maintenance of non-mtu components

Please refer to the Operating Instructions provided by the relevant component manufacturer for the intervals for these maintenance tasks.

Maintenance tasks

The respective maintenance tasks to be carried out are based on operating hours (see the following pages).

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1.4 Checks by the operator

Important
Always observe the safety instructions.

Important
Non-observance of these instructions may lead to damage to the system!

Important
When the object of delivery is used abroad or in other special regions, Rolls-Royce Solutions is not responsible for observance of the statutory and other specifications / standards applicable at the place of use.

Recommended daily checks – System

- Check exhaust pipework for condensate leaks (visual inspection)
- Check condensate collecting tank after condensate drain (if fitted)
- Carry out visual inspection of engine and plant for leaks / damage (bellows, hoses etc.)
- Check gas supply / gas system for damage (visual inspection)

Recommended daily checks – Control system

The following measured values and functions are monitored by the control system to protect the plant. These messages can also be called up via remote access (option) and can be monitored by the customer on the display.

- Electrical output
- Lube oil level
- Lube oil pressure
- Coolant temperature
- Lambda probe voltage (option)
- Battery voltage
- Smooth running of engine
- Exhaust temperature
- Heating water supply and return temperature (option)
- Oil bleed system function (option)

Recommended weekly checks

- Lube oil consumption or content of fresh oil tank
- Engine room temperature
- Check of operating hours
- Check of number of starts
- Check of speed control for stability of regulator and the linkage (if fitted)
- General visual inspection of module / engine-generator set
- Visual inspection of heat insulation of SCR system (if fitted)
- Visual inspection of tanks, lines and dosing system of SCR system (if fitted)
- Check of reducing agent concentration of SCR system (if fitted)

Additional checks

The operator's gas installation (upstream of twin solenoid valve) must be inspected for leaks at the following intervals:

- Every year for engines with turbocharger
- Every six months for naturally-aspirated engines

The following checks are to be carried out at appropriate intervals, depending on the degree of contamination:

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- Check room air fan and fresh air filter for contamination
- Check filter or air-conditioning unit for control cabinet for contamination
- Check return-flow cooler for contamination

Electrical systems

Electrical systems and operating equipment must be checked prior to initial commissioning, after repairs, prior to re-commissioning and at regular intervals. The intervals must be arranged such that defects, which are to be expected, are detected at an early stage.

Here, among others, the following requirements apply:

- Accident prevention regulations BGV A3 "Electrical Systems and Operating Equipment"
- Technical rules for operational safety TRBS 2131 "Electrical Hazards"
- Industrial Safety Directive (BetrSichV)
- DIN VDE 0701/0702 Repairs, modifications and testing of electrical equipment/repeat tests on electrical units
- DIN EN 60204-1 Safety of machinery - Electrical equipment of machines

The operating company has the responsibility to see that these tests are carried out correctly. Electrical systems and fixed electrical operating equipment must be tested at appropriate intervals by a qualified electrician for their correct condition. The test is to include the following:

- Visual inspection and testing of protective equipment
- Testing of protective conductor resistance, insulation resistance, leakage current replacement, touch current, protective conductor current and differential current
- Carrying out fault diagnoses
- Preparation of test reports and inspection tags

The industrial safety regulations, the technical rules for operational safety and the BGV A3 are fundamental with regard to adherence to the protective measures for "Electrical Systems and Operating Equipment". The operating company is accordingly obligated to care for the safety of electrical systems and operating equipment.

	Designation
DIN	Deutsches Institut für Normung (Federal German Standards Institute)
EN	Europäische Normung (European Standards)
ISO	International standard
VDE	Verband der Elektrotechnik, Elektronik und Informationstechnik (Association for electrical engineering, electronics and information technology)
BGV	Berufsgenossenschaftliche Vorschriften (Trade association regulations)
TRBS	Technische Regeln für Betriebssicherheit (Technical rules for operational safety)

Table 2: Designations for standards and regulations

Important
These regulations are contained in the manufacturers' instructions, statutory regulations and technical guidelines valid in the individual countries. Since there can be major differences from country to country, it is not possible to provide a universally applicable statement on the rules to be observed within the framework of this maintenance schedule. The user of the product mentioned herein is obligated to inform him-/herself about the provisions applicable in his/her country. Rolls-Royce Solutions accepts no liability whatsoever for improper or illegal use of the electrical system, which it has approved.

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1.5 Non-periodic maintenance tasks

No.	Quali-fica-tion	Task	Maintenance tasks	value 1	value 2	Unit
0001	QL1	W1689	Measure valve protrusion	250	32250	HRS
0002	QL1	W1002	Check valve clearance; adjust if necessary	250	32250	HRS
0003	QL1	W1008	Replace engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years)	250	32250	HRS
Note: The non-periodic maintenance tasks listed below must be carried out in addition to the recurring, periodic maintenance tasks at the specified runtime in operating hours. Value: Fixed value for performance of the maintenance task.						

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1.6 Maintenance schedule matrix

0 to 58,000 Operating hours

Task	Limit	Operating hours [h]																												
		Daily	2,000	4,000	6,000	8,000	10,000	12,000	14,000	16,000	18,000	20,000	22,000	24,000	26,000	28,000	30,000	32,000	34,000	36,000	38,000	40,000	42,000	44,000	46,000	48,000	50,000	52,000	54,000	56,000	58,000
Engine-generator set operation																															
W0744	1 a																														
W0742	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0743	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0823	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0833	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0847	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0861	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0864	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0873	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0740	4 m																														
Engine																															
W0509	-	X																													
W0741	1 a																														
W0868	1 a																														
W0500	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0710	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0711	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0727	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0728	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0730	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0731	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0734	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0737	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1923	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W9018	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0725	3 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W4175	3 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0732	18 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0733	18 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1002	18 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1091	18 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1168	18 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W0716	3 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W0719	3 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1689	18 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1089	3 a				X			X			X			X			X			X			X			X			X		
W0869	6 a				X			X			X			X			X			X			X			X			X		
W4172	1 a					X				X				X				X				X				X				X	
W0726	3 a					X				X				X				X				X				X				X	
W0713	18 a					X				X				X				X				X				X				X	
W1011	18 a					X				X				X				X				X				X				X	

w = weeks
m = months
a = years

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